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Baseball Salaries and Income Taxes: The “Home Field Advantage” of Income Taxes on Free Agent Salaries

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In this paper we examine the impact on the salaries of free agents in Major League Baseball of differences in state and local individual income taxes between major league cities, in an attempt to see if income taxes affect player salaries. Our basic specification suggests that each percentage point of an income tax raises free agent salaries by \$21 to \$24 thousand; other estimates indicate even larger impacts. Our findings suggest that the existence of this additional salary demand means that low tax cities (e.g., Florida, Texas, and Washington) have a “home field advantage” in the baseball free agent market.

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In this paper we examine the impact on the salaries of free agents in Major League Baseball of differences in state and local individual income taxes between major league cities, in an attempt to see if income taxes affect player salaries. Our basic specification suggests that each percentage point of an income tax raises free agent salaries by \$21 to \$24 thousand; other estimates indicate even larger impacts. Our findings suggest that the existence of this additional salary demand means that low tax cities (e.g., Florida, Texas, and Washington) have a “home field advantage” in the baseball free agent market.

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“We have no state taxes, and it’s always 80 degrees.” Doc Rivers, then-coach of Florida’s Orlando Magic, yelling from the sidelines to prospective free agents on opposing teams.

Introduction

The theory of tax incidence suggests that taxes will fall on fixed factors of production, while mobile factors will largely be able to avoid bearing the burden of a tax. It is hard to imagine many factors of production more mobile than free agent professional athletes about to sign a new contract. As such, an expectation about those contracts is that there will be a divergence between the tax inclusive payment made by a team and the tax exclusive payment received by the players. Assuming that free agents compare after-tax bids from prospective teams, offers that seem nominally identical will differ in their face value by the degree to which state (and local) taxes, especially state individual income taxes, differ in the playing sites.

The issue of taxation of professional athletes is particularly important in Major League Baseball (MLB). For example, in 2001 the Seattle Mariners’ Aaron Sele and the Detroit Tigers’ Dean Palmer both had a contractual, pre-tax salary of \$7,500,000. However, Palmer’s salary was reduced by \$315,000 from the State of Michigan’s income tax and by an additional \$93,750 from the City of Detroit’s income tax, reducing his pre-tax salary by \$408,750 (assuming individual filing). In contrast, Sele was able to keep the entire \$7.5 million since neither Washington nor Seattle imposes income taxes. However, if Sele had been traded to the New York Yankees (at the same salary), he would have owed even more than Palmer (\$801,750) in state and local taxes. As another especially telling example – and one based on actual events – during the winter between the 2002 and 2003 seasons, a trade between Florida, Colorado, and Atlanta involving several players and about \$200 million in contracts almost fell through in its

final stages when one player (Charles Johnson) refused to void a no-trade clause in his contract until he was compensated by an additional \$1,000,000 to offset additional tax liability due to his move from Miami (with no Florida individual income tax) to Colorado (where an income tax of nearly 5 percent was imposed). Differing rates of state and local income taxes can therefore cause significantly different after-tax salary results. It seems likely that free agent pre-tax salaries will reflect these differences in income taxes. However, despite the large and growing literature on the determinants of individual baseball player salaries, it is striking that (to our knowledge) there is no empirical evidence on this issue. There is also little empirical evidence from other professional sports in which the same considerations most likely arise (e.g., football, basketball, hockey).¹

State and local practices in income tax make the issue of taxing professional athletes of significant importance. All states with an individual income tax reserve the right to tax any non-resident on professional income earned in the state. This policy applies to any non-resident, but it is typically applied mainly to high-profile – and high-income – professional athletes like MLB players.² The most commonly used allocation method is based on “duty days”, or the number of days that the player spends in providing professional services in a state; the total salary of the player is then allocated across states in which he plays in accordance with the proportion of the total duty days spent in each state, and the player is required to file an individual income tax return in all states in which he plays and in which an income tax is imposed, either at the state or

¹ A recent exception is Kopkin (2011), who finds that state and local income tax rates affect the “migration” of National Basketball Association (NBA) free agents from one location to another. Note that much empirical work on the impact of marginal tax rates on the compensation of high income individuals indicates a significant behavioral response. For example, see Goolsbee (2000), Alm and Wallace (2000), and Bakija and Slemrod (2004). For a contrary view, see Young and Varner (2011).

² Some states have reciprocity agreements with other states, in which taxes paid to one state can be credited against the tax liability of another state. However, despite these reciprocal agreements, there is often double taxation of professional athletes. See Ekmekjian (1994), Ringle (1995), Green (1998), Barger (1999), Hawkins, Slay, and Wallace (2002), and Hoffman (2002) for detailed discussions of state and local income tax practices.

the local level. A player who is signed by a state with no income tax (e.g., Florida, Texas, Washington) will have most of his duty days untaxable, even though days in other states with an income tax will be subject to the tax. MLB players can often be required to file tax returns in upwards of a dozen or more states.

In this paper, we examine whether free agent baseball contracts reflect, or incorporate, differences in state and local individual income taxes. We collect annual salary and performance data on all 372 free agent players (171 pitchers and 201 position players) over the period 1995 to 2001, or the period after the cancellation of the 1994 season and through the expiration in 2001 of the Collective Bargaining Agreement between the major league baseball clubs and the players association. We use these data to estimate a within-group panel model of individual baseball free agent player salaries, estimated separately for pitchers and position players, as a function of individual performance indicators and individual fixed and time effects. Importantly, we include a variable that measures the marginal tax rate in state and local individual income taxes (where relevant), in order to examine how income taxes affect player salaries. We find in our basic specification that individuals choosing to play in cities with income taxes must be paid higher pre-tax salaries, by an amount that ranges from \$21 to \$24 thousand for each percentage point of a state and local income tax; other estimates suggest even larger impacts. We then point out the implications for sports salary modeling, as well as the implications for major league baseball from state and local tax differences, differences that are presumably unintended and certainly outside the control of baseball. Our basic conclusion is that free agents are mobile factors of production, and, as a result, they bear little of the burden of individual income taxes.

Modeling the Determinants of Baseball Player Salaries

Because so much information is available relating to player productivity in baseball, a rich literature has developed investigating the relationship between the value of player performance and team revenue generation (e.g., a player's marginal revenue product, or MRP). A major issue in this literature has been the degree to which MRP and salary tend to diverge as a result of monopsony power on the part of teams. Prominent in this literature are contributions by Scully (1974), Zimbalist (1992), and Fort (1992). More recently, Krautmann (1999) has demonstrated that the existence of an open and competitive market for players eligible for free agency allows salaries for free agents to be more directly tied to relevant productivity factors. See also recent empirical studies by Bradbury (2007), Healy (2008), Brown and Jepsen (2009), and Krautmann and Solow (2009).

At question for our work here is which of the many player productivity variables might best predict salary. Everyday position players have both offensive and defensive characteristics, with some degree of differentiation among various attributes. The important offensive contributions of, say, a lead-off hitter, a number two hitter, or a clean-up hitter will vary, just as the important defensive characteristics of catchers, infielders, and outfielders are not the same. Nevertheless, most researchers have found that there exists a wide range of variables that are correlated with salaries and, indeed, that simple models are often as effective as complicated ones (Fort, 1992). On the other hand, empirical models for pitchers tend not to be as successful in explaining pitcher salaries, given the apparent difficulties in aggregating over different types of pitching measures (Krautmann, Gustafson, and Hadley, 2003; Bradbury, 2007)). Underlying all of these models is the presumption that past performance predicts future salary. Since it is future salary and not future performance that is being predicted, past performance seems a reasonable indicator.

What role do state income taxes play in determining player salaries? The theory of tax incidence suggests that taxes will be shifted from mobile factors to fixed factors (Fullerton and Metcalf, 2002). Since free agent baseball players are quite mobile, certainly more mobile than the baseball franchises for whom the players work, this theory implies that franchises will bear the bulk of the burden of state income tax differences. However, to the extent that franchises are themselves footloose, as in the case of new franchises or relocations of existing franchises, then local governments compete extensively through tax breaks to win franchises (Noll and Zimbalist, 1997). In any event, free agents cannot be considered fixed factors, and so they should effectively escape the incidence of state income taxes. Consequently, we argue that players will compare after-tax salary offers, whereas reported salaries – and those salaries tested in all other baseball salary models to date – are invariably pre-tax salaries.³ To the extent that these two figures differ by any considerable amount, there is a clear misspecification of the salary models.

Do State and Local Taxes Differ Across Baseball Sites?

An important, prior empirical issue is whether state and local taxes differ to any significant degree across team sites. The answer to this question is “Yes”. Five of the thirty MLB teams are in sites with no state or local income tax: Miami, Tampa Bay, Houston, Texas, and Seattle. Table 1 lists the highest state income tax bracket by state with a baseball franchise, in 2001 (or the last year of our sample); for purposes of comparison, Table 1 also presents similar information as if 1 January 2011. In addition, local income taxes of up to 4 percent may be applied in some cities, with Baltimore (3.05 percent), Cincinnati (2.10 percent), Cleveland (2.0 percent), Detroit (1.25 percent), Kansas City (1.0 percent), New York City (3.84 percent), Philadelphia (3.9642 percent), Pittsburgh (1.0 percent), San Francisco (1.5 percent), and St.

³ See Solow and Krautmann (2011) for a game theory model of free agent salary determination.

Louis (1.0 percent) all having local rates at or above 1 percent in 2001.⁴ This information suggests a variability of up to about 10 percent in top marginal tax income tax rates in U.S cities alone. When franchises in Canada are considered, the sum of the highest provincial and national marginal rates in Canada in 2001 was 40.16 percent for Ontario and 53.5 percent for Quebec, thereby creating an even greater variation in top marginal tax rates.⁵

Professional athletes, and multiple site performers of any kind, are responsible for paying taxes to state and local authorities at every venue at which they play. Under current inter-league playing schedules, a player on a major league roster for a full year may play in up to 18 or 19 different cities scattered over a dozen or more states. For instance, in 2001 a member of the Colorado Rockies on the major league roster for a full year played in 18 cities over 14 states and one province, and was responsible for paying the applicable state and local tax in each location unless some sort of reciprocal agreement exists between the states. Playing half of one's games in a high relative to a low tax venue can make a considerable difference on a player's after-tax income. Recall the earlier example of Charles Johnson. Moving 81 home games from a city with no state or local income taxes to one with a state tax rate of 4.63 percent prompted him to demand an additional \$1 million in compensation.⁶ It is this issue that we examine next.

Empirical Framework

⁴ These city tax rates are based on our search of individual city websites.

⁵ In the period that we examine, there was a MLB franchise in Montreal, Quebec, the Montreal Expos. This franchise has since been relocated to Washington, D.C., and renamed the Washington Nationals. There remains one MLB franchise in Canada, the Toronto Blue Jays.

⁶ Again, see Ekmekjian (1994), Ringle (1995), Green (1998), Barger (1999), Hawkins, Slay, and Wallace (2002), and Hoffman (2002) for detailed discussions of state and local income tax practices.

Methods

We collect annual salary and performance data for all 372 baseball free agents (171 pitchers and 201 non-pitchers) for the period 1995 to 2001, which allows us to use variation across free agent salaries to identify and to investigate how the presence and the magnitude of state and local income taxes affects these salaries. We estimate a within-group model that exploits the panel nature of our data and that controls for individual fixed and time effects.⁷ Aside from the marginal tax rate variable, we also include a full array of control variables, including major league experience and other standard performance indicators for both pitchers and non-pitchers. We choose this period because it covers the entire period after the cancellation of the 1994 season and through the expiration in 2001 of the Collective Bargaining Agreement between the major league baseball clubs and the players association.

Denote S_{it} as the pre-tax salary of free agent i in period t . We assume that the basic relationship between the explanatory variables and the free agent salary is given by:

$$S_{it} = \alpha MTR_{it} + \beta X_{it} + \mu_i + \eta_t + \varepsilon_{it}, \quad (1)$$

(plus a constant term), where MTR_{it} is a variable that measures the sum of state and local marginal tax rates for free agent i in period t , X_{it} is a vector of performance indicators that determine the salary of free agent i in period t , μ_i and η_t are individual fixed and annual time effects, (α, β) are the relevant coefficients on the marginal tax rate and the performance indicators, and ε_{it} is a random error term. Note that free agents playing in a state for which there is no state or local income tax have a value of MTR_{it} equal to 0. We follow the previous literature by including a very wide range of performance indicators in numerous specifications,

⁷ Individual fixed effects capture any permanent differences across free agents (e.g., innate abilities) not otherwise captured by other explanatory variables. Similarly, the time effects capture any variation in free agent salaries over time that affects the whole country (e.g., changes in MLB franchise laws, federal income taxes).

and we also explore alternative functional form specifications (e.g., quadratic, log-linear, log-log), all in an attempt to demonstrate the robustness of our results.

The fixed-effects model seems appropriate for our analysis for two reasons. First, much of the variation in free agent salaries is between individuals rather than within the same individual over time. Although it would be difficult to specify all the characteristics that determine the differences across individuals in free agent salaries, we can capture permanent differences between free agents with individual fixed-effects. Similarly, there are many factors that may affect free agent salaries over time, and we capture those differences with annual time effects. Second, the fixed-effects model is a within-group estimator that uses a weighted average of the within-individual and the across-individual variation to form the parameter estimates. Therefore, our estimate of the effects of state and local income tax variations measures how free agent salaries change within panels of free agents due to the presence or absence of a state and local income tax. Nonetheless, we also estimate similar specifications with a random-effects model, and our results are largely unaffected.⁸ Results from the Hausman (1978) specification test indicate that the fixed effects model is a better fit. All standard errors are estimated using the White (1980) procedure for robust standard errors.

Data

Our dependent variable is the annual pre-tax salary of baseball free agent i for time period t , measured in thousands of dollars. We obtain information on free agent salaries and other performance indicators for the years 1995 through 2001 from the MLB website, for all MLB clubs. We include several explanatory variables to measure the effect of state income taxes on salaries across free agents and over time. Our primary regressor is the top marginal tax

⁸ Hsiao (1986) presents a comprehensive discussion of panel data estimation procedures.

rate in the state and local income tax (where imposed); given that the top marginal tax rate applies to incomes well below the salary of free agents, there is little concern with potential endogeneity of the MTR_{it} variable.⁹ Our expectation is that the estimated coefficient on MTR_{it} will be positive; that is, a player signing a contract in a city with a state and/or a local income tax will require that the pre-tax salary of the free agent be higher in order to compensate for the added income tax liability, in an amount that depends on the presence and on the magnitude of the income tax in the city and the state. We estimate several variants on the basic specification in equation (1), including those that are quadratic in MTR_{it} , that are in log-linear form (e.g., S_{it} is measured in natural log form), and that are in log-log form (e.g., both S_{it} and MTR_{it} are in log form).

To assess the impact of state income taxes on free agent salaries, it is necessary to control for other factors that potentially affect these salaries. Following Krautmann, Gustafson, and Hadley (2003), we include several factors that measure the performance (annual or career) of hitters and pitchers. For position players (or non-pitchers), these performance and other measures include: major league experience (*Experience*) and major league experience squared (*Experience Squared*), on base percentage plus slugging percentage (*OPS*), and fielding percentage (*Fielding Percentage*). Salary determinants for the pitchers include: major league experience and major league experience squared, wins (*Wins*), the win/loss average (*WinLoss Average*, computed as the ratio of wins to the sum of wins and losses), innings pitched (*Innings Pitched*), earned run average (*ERA*), saves (*Saves*), and strike-out-to walk ratio (*KBB Ratio*). For both non-pitchers and pitchers, we estimated different specifications with previous season and with career averages of these performance indicators to appropriately capture their effect on free agent salaries. Further, all specifications include year dummies to capture the trend of free agent

⁹ For example, in 2001 the average MLB salary was \$2.2 million, and the minimum salary was \$200,000.

salaries during the period 1995-2001. Table 2 summarizes the descriptive statistics for our main variables. Annual performance indicators in the most recent year before free agency have the suffix *A*, while the career indicators are denoted by the suffix *C*.

It should be noted that we have estimated specifications with a wide range of alternative performance indicators. For pitchers we have estimated variants with annual and career indicators included and excluded, a dummy variable equal to 1 if the pitcher is left-handed and 0 otherwise, and so on. For non-pitchers, we have included annual and career batting average, annual and career runs batted in (*RBI*), annual and career home runs, and the like. We have also experimented with some of the more recently devised sabermetrics statistics for non-pitchers and pitchers, such as “base runs” or “runs created”, “secondary average”, “true average”, “range factor”, “weighted on-base average”, “equivalent average”, “peripheral earned run average”, “defense-independent earned run average”, “walks plus hits per innings pitched”, “win shares”, “total player rating”, and “value over replacement player”, to name several.¹⁰ These alternative measures sometimes affect the magnitude of the tax impact on salaries but not its sign. We report the results from the more commonly used performance statistics, but all results are available upon request.

Estimation Results

Table 3 contains the regression results of free agent salary estimation for non-pitchers for the years 1995 through 2001. We report findings the random and fixed effects models for

¹⁰ See James (1989, 2001, 2011) and Thorn, Palmer, and Gershman (2004) for more detailed discussion of these methods. For detailed discussions of the many statistics, see especially the website of the Society for American Baseball Research (or “sabermetrics”) at <http://www.sabr.org/>, the Baseball Prospectus at <http://www.baseballprospectus.com/>, or The Hard Times at <http://www.hardballtimes.com/>.

comparison. These results are largely the same, and our discussion is based mainly on results from the fixed effects model.

For non-pitchers, the presence of a state and local individual income tax raises salaries by about \$24 thousand for each percentage point of the income tax. Also, most of our explanatory variables used to measure the effect of free agents' performance on salaries have the expected signs. For instance, an additional year of MLB experience initially leads to an increase in a free agents salary by \$242 thousand while MLB experience beyond six years lowers free agent salaries for non-pitchers by \$15 thousand. Also, an increase in the previous season's on-base plus slugging percentage raises a free agent's salary by approximately \$1.1million, and an increase in the lifetime on-base plus slugging percentage also has a comparably large and statistically significant effect on free agent salaries for non-pitchers. In contrast, lifetime and previous season fielding percentages do not have a statistically significant effect on free agent salaries. The year dummies suggest that there was no statistically significant difference in free agent salaries by year during the period 1995-2001.

Table 4 shows the estimation results from the free agent pitcher salary equation. These findings indicate that state and local income taxes again have a positive and statistically significant effect on salaries for free agent pitchers, with a slightly smaller impact than for non-pitchers. The presence of a state and local income tax now raises pitcher salaries by \$21 thousand for each percentage point of the income tax; the random effects results gives a largely similar impact. Unlike the results for non-pitchers, experience does not have a significant impact on free agent pitcher salaries. As for performance measures, annual and lifetime wins both have a positive impact on salaries, as does previous season innings pitched, previous season win-loss average, the career strikeout-walk ratio, and the previous season saves total. As in the non-

pitchers salary equation, the year dummies suggest that there is no statistically significant difference in free agent salaries for pitchers by year over most of the period 1995-2001, with the exception of the last two years.

To test further the robustness of these results, Table 5 presents fixed effects models for both non-pitchers and pitchers that examine the impact of different ways of introducing MTR_{it} . Variant 1 repeats the fixed effects results for non-pitchers and pitchers as reported in Tables 3 and 4. Variant 2 introduces the marginal tax rate in a quadratic form (or MTR plus MTR Squared). Variant 3 uses a simple log-linear specification where S_{it} is measured in its natural log form, and Variant 4 estimates a standard log-log specification. Other variables are entered in the same way as in the basic fixed effects specifications of Tables 3 and 4.

These results indicate somewhat variable quantitative impacts of the state and local income tax on free agent salaries. Even so, all variants clearly indicate that a state and local income tax has a positive and significant qualitative impact on free agent salaries. For example, the quadratic specification suggests that the dollar impact on salaries of the income tax increases (not decreases) with an increase in the marginal tax rate. Similarly, the log-linear variant indicates that the percentage response of salaries to an increase in the marginal tax rate is about 2 to 3 percent. The log-log specification shows an elasticity of salaries with respect to the marginal tax rate slightly in excess of 0.5. All variants point to the importance of state and local income taxes in free agent salary determination.

Conclusions and Implications

There is a wide range of state and local income tax rates among cities with baseball franchises. This range generates a marginal rate difference of up to 10 percentage points

between those cities in the U.S. with an income tax and the five no-tax cities. Our estimation results indicate that these tax differences are largely reflected in pre-tax salary offers to free agent MLB players: free agents are highly mobile, and they are able to ask for and receive pre-tax salaries that largely compensate them for the state and local income taxes that they must pay if playing in cities with an income tax.

To illustrate, the average salary of a MLB player in 2001 was \$2.2 million. If the average top marginal tax rate in state and local income taxes is approximated by 7 percent, as indicated in the descriptive statistics of Table 2, then a player choosing the free agent offer of a city with an income tax over a city without one would incur an additional statutory income tax liability of roughly \$150,000.¹¹ Our estimation results vary somewhat in their precise implications, but even so our basic specification suggests that each percentage point of a state and local income tax raises free agent salaries by \$21 to \$24 thousand, a result that is broadly similar to the increased statutory tax liability; other variants give largely similar estimates. In short, income taxes are largely shifted away from mobile players to other factors, such as the franchise itself or immobile factors in the city.

There are other implications of our results as well. The total 2001 payroll for all MLB teams was in total \$1.9 billion, of which \$1.6 billion was incurred by franchises operating in states with a state and local income tax. Again assuming that average top marginal tax rate is roughly 7 percent, the franchises in states with an income tax paid an additional \$20 million to compensate players for the additional income taxes to which their players were subject.

¹¹ Note also that in 2001 three-fourths of all states with an individual income tax imposed a top marginal tax rate of between 5 and 9 percent; local income taxes impose additional burdens. See Table 1; for other years, see the Tax Foundation information available at http://www.taxfoundation.org/files/state_individualincome_rates-2000-2011-20110302.pdf.

An examination of individual players is also instructive. In 2001, there were 425 players who were paid more than \$1 million, and 22 players who were paid more than \$10 million. The five highest paid free agent players were Alex Rodriguez (Texas Rangers, \$22,000,000), Kevin Brown (Los Angeles Dodgers, \$15,714,286), Mike Piazza (New York Mets, \$13,571,429), Randy Johnson (Arizona Diamondbacks, \$13,350,000), and Mo Vaughn (Anaheim Angels, \$13,166,667).¹² Of these five top earners, the salaries of all – notably except Rodriguez – were subject to significant state and local income taxes that ranged from about 5 percent (for Johnson) to roughly 10 percent (for Brown, Piazza, and Vaughn). Our estimation results suggest that the ownership of the teams in states with income taxes had to increase significantly the pre-tax salaries of these players in order to sign them. In contrast, the Texas Rangers got a relative bargain in Rodriguez given the absence of any state/local income tax in Texas. Similar adjustments were necessary for other free agent players.

The basic implication of this tax difference is that there is a competitive edge – a “home field advantage” – for teams in low-tax areas because they have lower team expenses in signing free agents to contracts that pay the same after-tax wage to players. After correcting for multiple tax venues, Florida, Tampa Bay, Texas, Houston, and Seattle may be able to sign free agents at a salary savings of from 2 to 3 percent relative to other clubs.

This result leads to other implications as well. Because player contracts are assignable in MLB, clubs in low-tax cities will find it easier to trade players to new teams because those players will have relatively lower salaries. Trades in baseball are frequently motivated by a desire to “dump salary”. When the salaries of those players being dumped are lower to begin with, trades will be easier to negotiate and may bring better terms for low tax cities.

¹² See http://sportsillustrated.cnn.com/baseball/mlb/news/2001/04/04/millionaires_ap/.

On the other hand, in order to counter an abrupt and undesired change of teams, players commonly negotiate limited no-trade clauses in their contracts. Whereas these clauses are commonly thought of as restricting player movement to other clubs that tend to be more competitive or to clubs in areas with player-desired amenities, it is likely that an examination of clubs to which players with trade restrictions would allow themselves to be traded would reveal a disproportionate representation of low-tax cities.

Finally, the salary savings advantage for low-tax franchises will be magnified by the present “luxury tax” system among teams. In order to maintain competitive balance and slow the overall growth of salaries, MLB has instituted a system by which teams with the total annual payrolls above a “threshold” must pay an additional “luxury tax” on all payroll above the limit, with the tax payments then distributed to teams with the lowest payrolls. Since clubs in high tax cities find they must pay more for the same players, the luxury tax distorts in favor of teams in low tax cities. For example, in 2003 (the first year in which the luxury tax was imposed), the New York Yankees had player salaries that were \$67.5 million above the threshold of \$117 million. With a luxury tax rate in 2003 of 17.5 percent, the Yankees paid total taxes of \$11.2 million. Our estimation results suggest that as much as \$7 million of the above-threshold salaries was due to the existence of state and local income taxes, which implies that over \$1 million of the luxury tax was due to the income taxes. The likelihood that state and local income taxes contributed to higher player salaries exacerbated the penalty imposed on the Yankees.

In short, free agent salaries are affected at least in part by the existence of individual income taxes. Players choosing to play in high-tax cities demand – and receive – higher free agent salaries as compensation. This phenomenon should be recognized in studies of the

determinants of player salaries, as well as in the analysis of other aspects of the market for professional baseball players.¹³

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¹³ See Fort (2009) for a recent discussion of some of the difficulties of empirical research in sports economics.

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Table 1. Top Marginal Tax Rate in U.S. State Income Tax in 2001 and 2011

State	Rate (percent)		Number of Franchises, 2001
	2001	2011	
Arizona	5.04	4.54	1
California	9.30	10.30	5
Colorado	4.63	4.63	1
Florida	0	0	2
Georgia	6.00	6.0	1
Illinois	3.00	5.00	2
Maryland	4.85	5.50	1
Massachusetts	5.60	5.30	1
Michigan	4.20	4.35	1
Minnesota	7.85	7.85	1
Missouri	6.00	6.0	2
New York	6.85	8.97	2
Ohio	7.50	5.925	2
Pennsylvania	2.80	3.07	2
Texas	0	0	2
Washington	0	0	1
Wisconsin	6.75	7.75	1

Source: Tax Foundation, Inc., available online at

http://www.taxfoundation.org/files/state_individualincome_rates-2000-2011-20110302.pdf .

The 2001 tax rates are as of 1 January 2002; the 2011 tax rates are as of 1 January 2011.

Table 2. Descriptive Statistics

Variable	Mean	Standard Deviation
<i>Non-pitchers</i>		
Salary (\$)	1,730,420	1,747,765
MTR (percent)	6.48	2.61
Experience (years)	10.58	3.53
On Base Percentage-A	0.336	0.051
On Base Percentage-C	0.326	0.042
Slugging Percentage-A	0.403	0.098
Slugging Percentage-C	0.389	0.059
OPS-A	0.742	0.129
OPS-C	0.714	0.085
Batting Average-A	0.266	0.046
Batting Average-C	0.262	0.030
Home Runs-A	7.71	8.12
Home Runs-C	95.30	95.66
RBI-A	37.40	27.73
RBI-C	458.55	341.67
Fielding Percentage-A	0.978	0.023
Fielding Percentage-C	0.973	0.019
Observations-Years	360	---
Observations-Individuals	201	---
<i>Pitchers</i>		
Salary (\$)	2,337,967	2,120,534
MTR (percent)	6.53	2.39
Experience (years)	9.48	3.03
ERA-A	4.43	1.53
ERA-C	4.14	0.79
Wins-A	5.86	4.80
Wins-C	66.25	45.12
Win Loss Average-A	0.513	0.242
Win Loss Average-C	0.527	0.089
Innings Pitched-A	97.74	69.75
Innings Pitched-C	1103.77	692.49
KBB Ratio-A	1.96	0.75
KBB Ratio-C	2.11	1.38
Saves-A	3.70	8.88
Saves-C	42.45	92.55
Observations-Years	267	---
Observations-Individuals	171	---

“A” denotes annual measures; “C” denotes career measures.

Table 3: Non-Pitchers – Estimation Results

Variable	Random Effects	Fixed Effects
MTR	24.02** (2.43)	23.55** (2.61)
Experience	308.75** (2.47)	242.17** (2.66)
Experience Squared	-11.32** (2.36)	-15.33*** (2.96)
OPS-A	1208.23** (2.40)	1104.02* (1.69)
OPS-C	6151.27*** (5.02)	1491.11* (1.92)
Fielding Percentage-A	291.63 (0.69)	32.21 (0.91)
Fielding Percentage-C	716.52 (0.27)	3568.78 (1.35)
Year 1996	-171.26 (0.99)	-67.28 (0.37)
Year 1997	-125.77 (0.59)	-129.85 (0.36)
Year 1998	-296.82 (1.61)	-286.91 (0.58)
Year 1999	-283.22 (1.37)	-376.70 (0.36)
Year 2000	199.54 (1.63)	1460.01 (0.82)
Year 2001	-41.71 (0.46)	-371.67 (0.39)
Constant	-4799.28* (1.79)	-5279.35* (1.97)
Observations	360	360
Number of Individuals	201	201
R-squared	0.36	0.47

Robust standard errors are in parentheses.

* significant at 10%; ** significant at 5%; *** significant at 1%

Table 4: Pitchers – Estimation Results

Variable	Random Effects	Fixed Effects
MTR	19.77* (1.84)	21.36** (2.34)
Experience	-168.69 (1.19)	-272.13 (0.72)
Experience Squared	-8.50 (0.64)	7.75 (0.59)
ERA-A	189.81*** (3.11)	149.04** (2.41)
Wins-A	177.11*** (4.27)	117.188** (2.27)
Wins-C	126.20*** (6.85)	88.67*** (2.75)
WinLoss Average-A	-1117.91*** (2.71)	-1206.16* (1.82)
Innings Pitched-A	12.52 (0.20)	8.53* (1.99)
KBB Ratio-C	178.58* (1.88)	113.80* (1.94)
Saves-A	49.49*** (3.32)	53.16* (1.94)
Year 1996	-99.08 (0.08)	-101.11 (0.11)
Year 1997	-127.85 (0.38)	-149.23 (0.19)
Year 1998	-328.21 (0.99)	-256.19 (0.17)
Year 1999	454.64*** (3.03)	404.46 (0.23)
Year 2000	601.11*** (4.58)	423.01** (2.41)
Year 2001	128.45*** (5.41)	527.87 (2.14)*
Constant	2210.30** (2.09)	2417.25** (2.41)
Observations	267	267
Number of Individuals	171	171
R-squared	0.52	0.33

Robust standard errors are in parentheses.

* significant at 10%; ** significant at 5%; *** significant at 1%

Table 5: Sensitivity Analysis

Variant	Non-pitchers	Pitchers
<i>1. Linear Variant</i>		
MTR	23.55** (2.61)	21.36** (2.34)
<i>2. Quadratic Variant</i>		
MTR	4.79** (2.56)	3.94** (2.77)
MTR Squared	11.43* (1.88)	9.03** (2.49)
<i>3. Log-linear Variant</i>		
MTR	0.03* (1.79)	0.02** (2.46)
<i>4. Log-log Variant</i>		
Ln MTR	0.66** (2.93)	0.51** (2.31)

Coefficient estimates from a fixed effects regression are reported with robust standard errors in parentheses.

* significant at 10%; ** significant at 5%; *** significant at 1%